

Introduction to Digital Image Processing

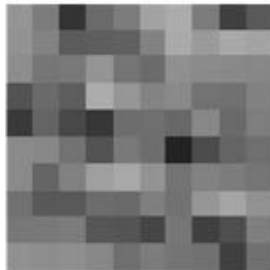
**PUSL3122 HCI, Graphics and
Visualization**



What is an Image?

- 2-dimensional matrix of Intensity (gray or color) values

Set of Intensity values $I(u, v) \in \mathbb{P}$ and $u, v \in \mathbb{N}$.
Image coordinates are integers



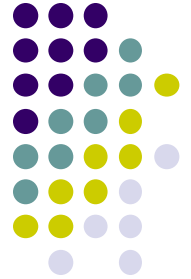
$F(x, y)$



148	123	52	107	123	162	172	123	64	89	...
147	130	92	95	98	130	171	155	169	163	...
141	118	121	148	117	107	144	137	136	134	...
82	106	93	172	149	131	138	114	113	129	...
57	101	72	54	109	111	104	135	106	125	...
138	135	114	82	121	110	34	76	101	111	...
138	102	128	159	168	147	116	129	124	117	...
113	89	89	109	106	126	114	150	164	145	...
120	121	123	87	85	70	119	64	79	127	...
145	141	143	134	111	124	117	113	64	112	...
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

$I(u, v)$

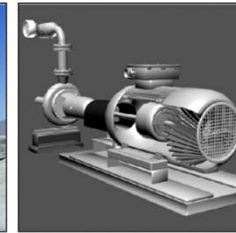
Example of Digital Images



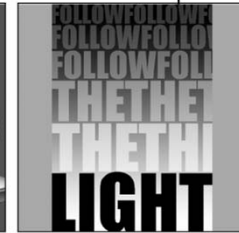
- a) Natural landscape
- b) Synthetically generated scene
- c) Poster graphic
- d) Computer screenshot
- e) Black and white illustration
- f) Barcode
- g) Fingerprint
- h) X-ray
- i) Microscope slide
- j) Satellite Image
- k) Radar image
- l) Astronomical object



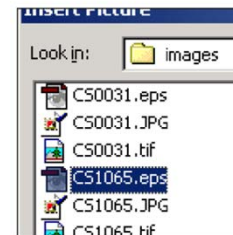
(a)



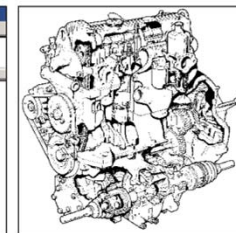
(b)



(c)



(d)



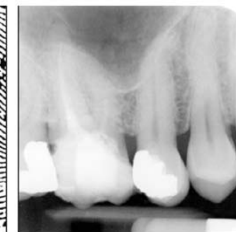
(e)



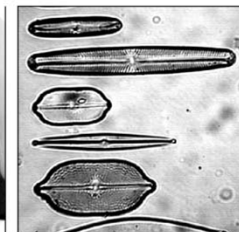
(f)



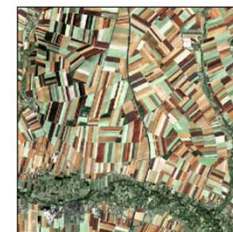
(g)



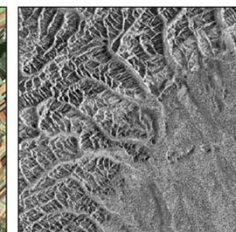
(h)



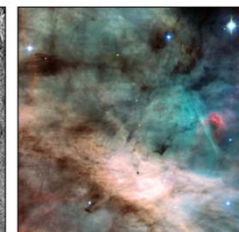
(i)



(j)

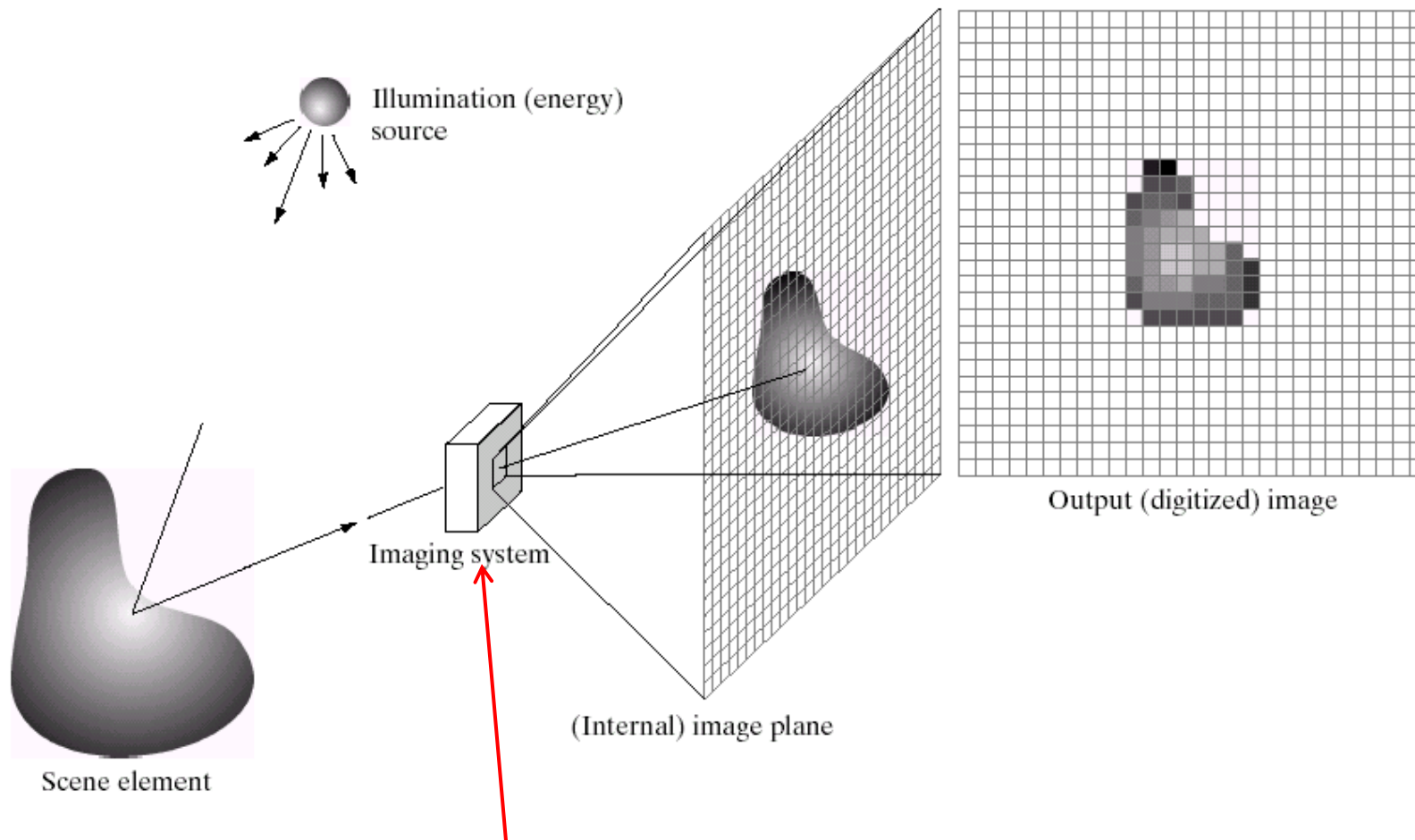


(k)



(l)

Imaging System



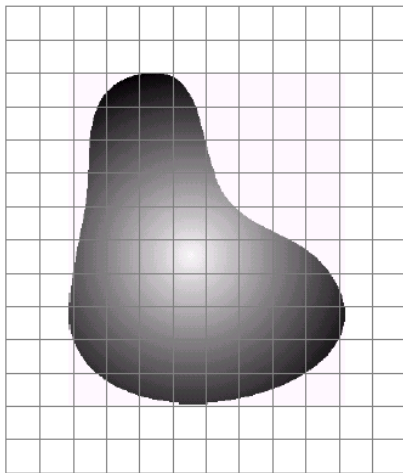
Example: a camera
Converts light to image

Credits: Gonzales and Woods

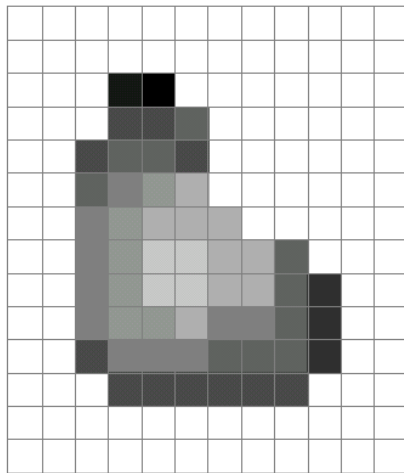


Digital Image?

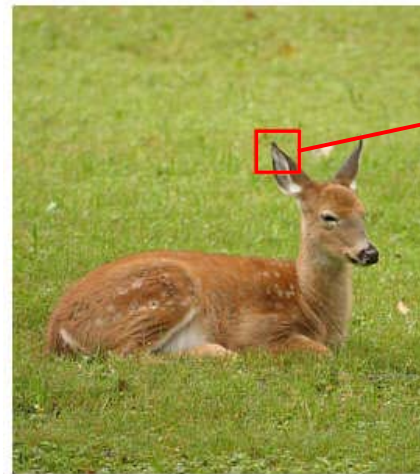
- **Remember:** *digitization* causes a digital image to become an *approximation* of a real scene



Real image



Digital Image
(an approximation)



Real image



Digital Image
(an approximation)



Digital Image

- Common image formats include:
 - 1 values per point/pixel (B&W or Grayscale)
 - 3 values per point/pixel (Red, Green, and Blue)
 - 4 values per point/pixel (Red, Green, Blue, + “Alpha” or Opacity)



Grayscale



RGB



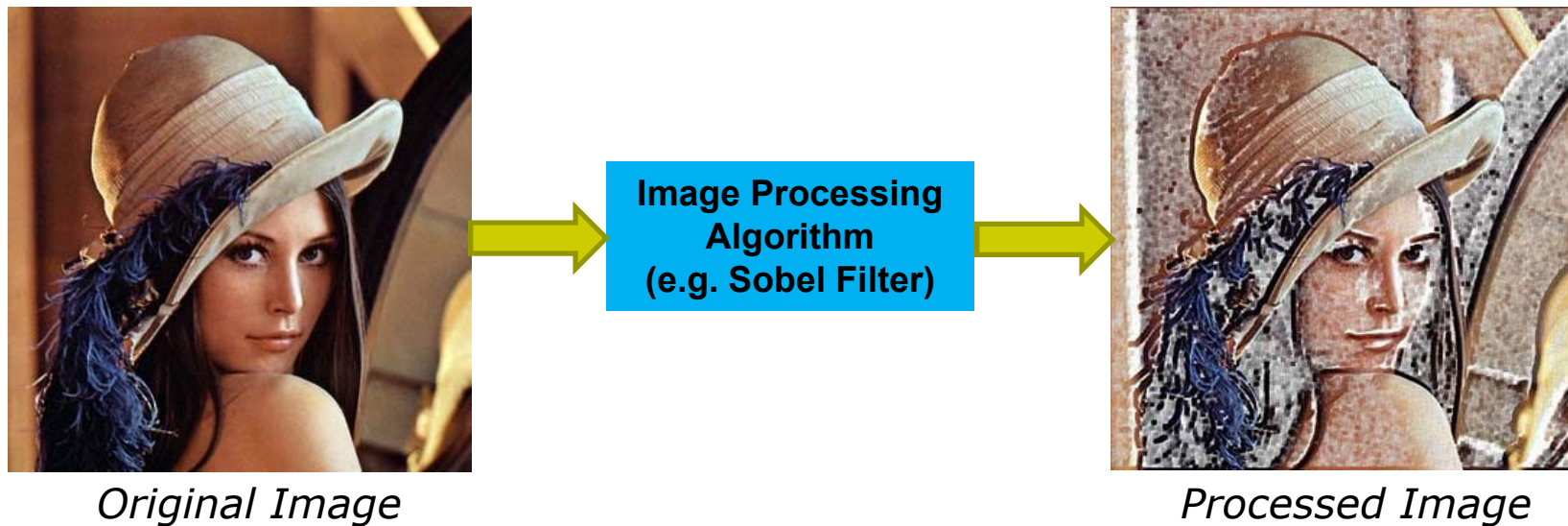
RGBA

- We will start with gray-scale images, extend to color later

What is image Processing?



- Algorithms that alter an input image to create new image
- Input is image, output is image

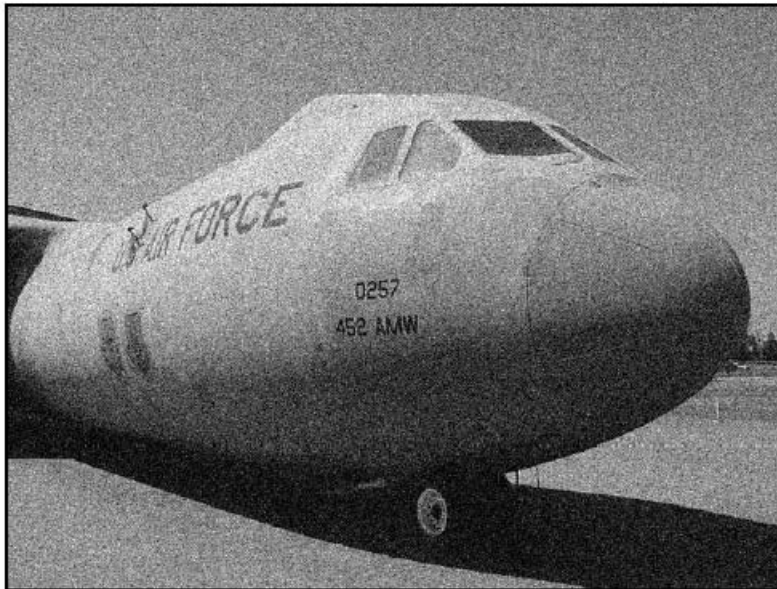


- **Improves an image for human interpretation in ways including:**
 - Image display and printing
 - Image editing
 - Image enhancement
 - Image compression



Example Operation: Noise Removal

Noisy Image

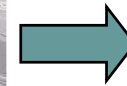
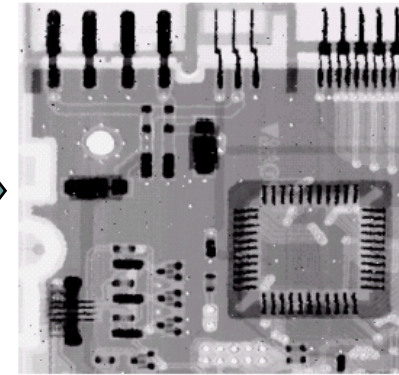
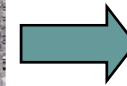
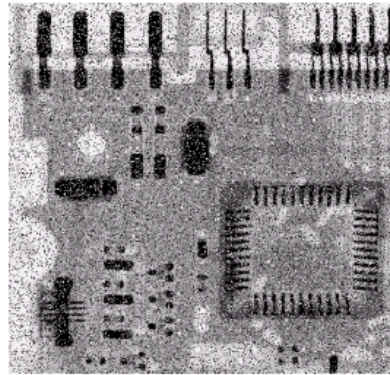
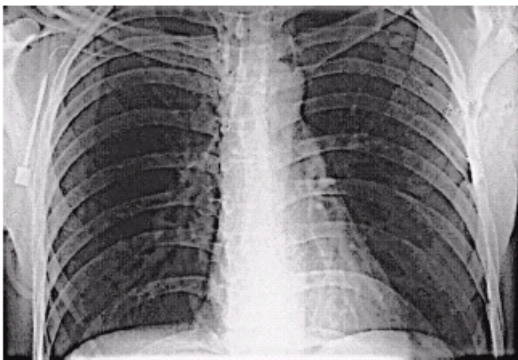
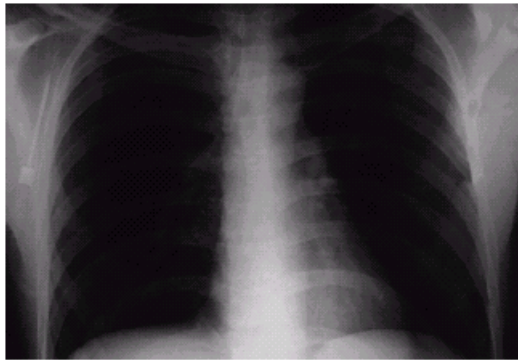
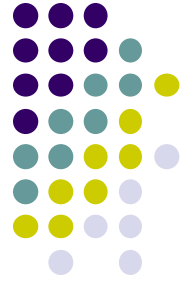


Denoised Image

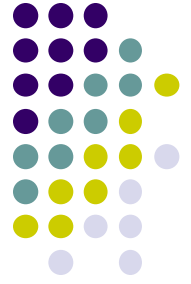


Think of noise as white specks on a picture (random or non-random)

Examples: Noise Removal



Example: Contrast Adjustment



Low Contrast

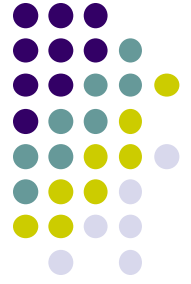


Original Contrast

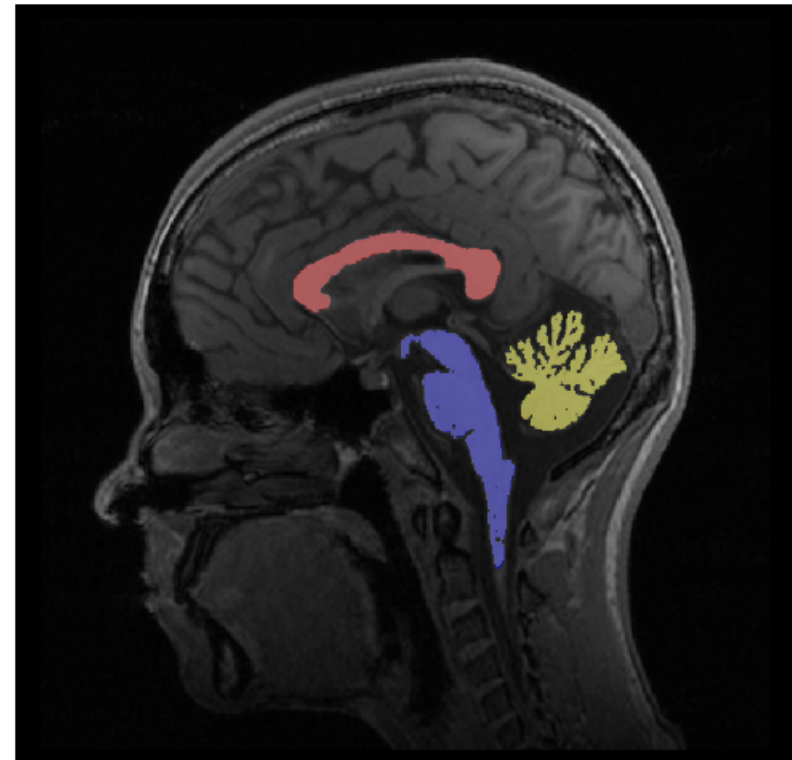
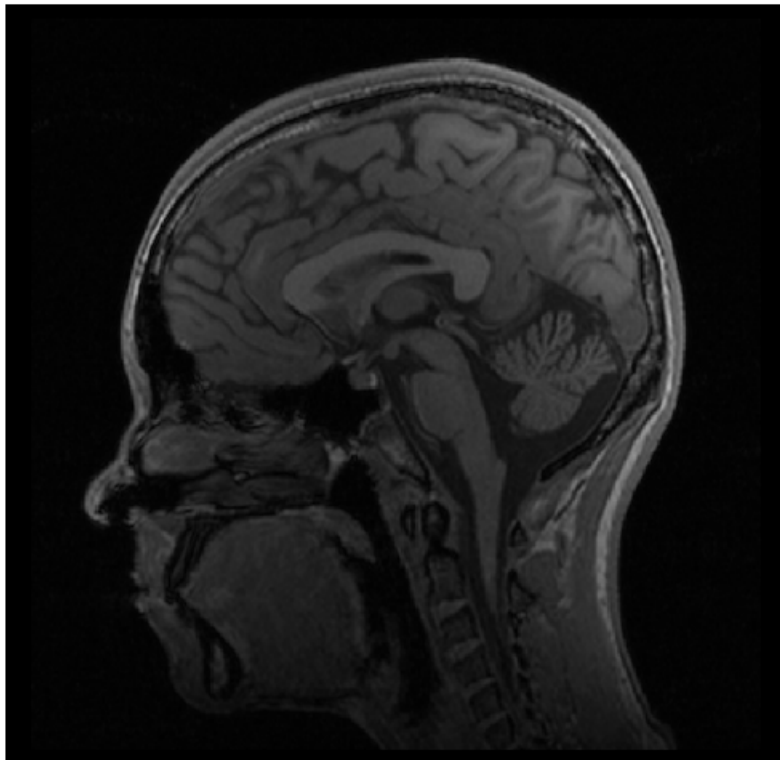


High Contrast

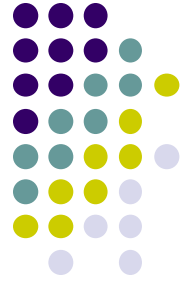
Example: Edge Detection



Example: Region Detection, Segmentation



Example: Image Compression



Original, 2.1MB



JPEG Compression, 308KB (15%)



Example: Image Inpainting

Damaged Image

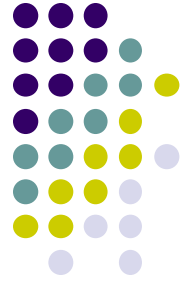


Restored Image



Credit: M. Bertalmio, G. Sapiro, V. Caselles, C. Ballester: Image Inpainting, SIGGRAPH 2000

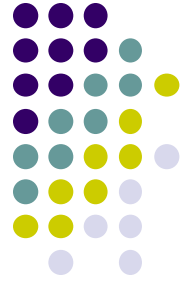
Inpainting? Reconstruct corrupted/destroyed parts of an image



Examples: Artistic (Movie Special)Effects

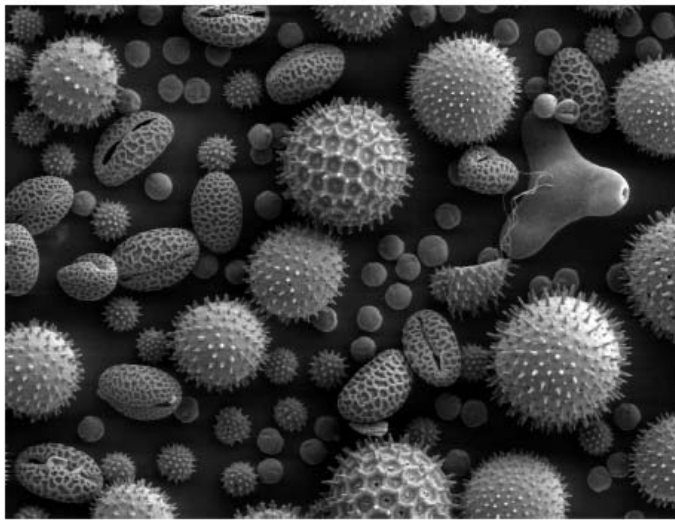


Applications of Image Processing

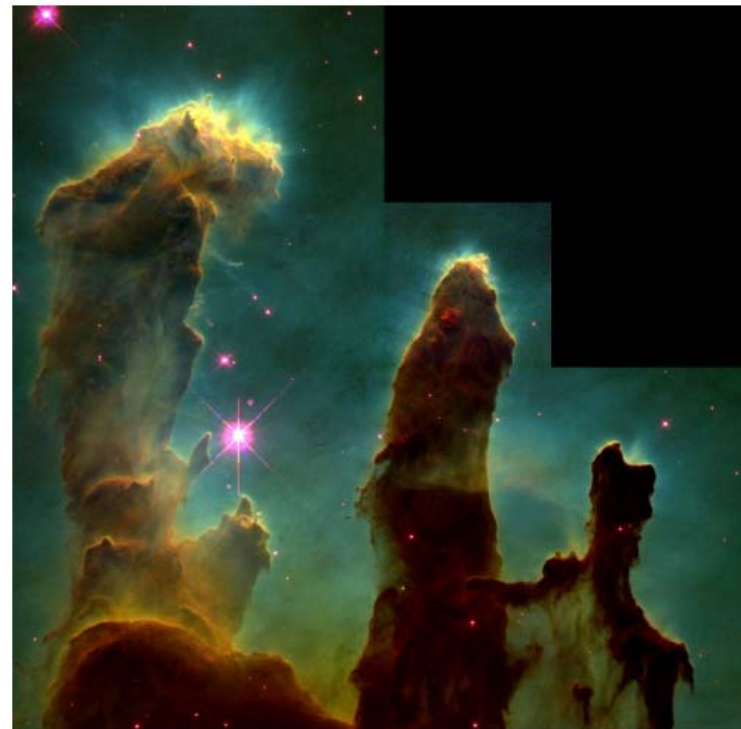


Astronomy

Biology



Credit: Dartmouth Electron Microscopy Facility

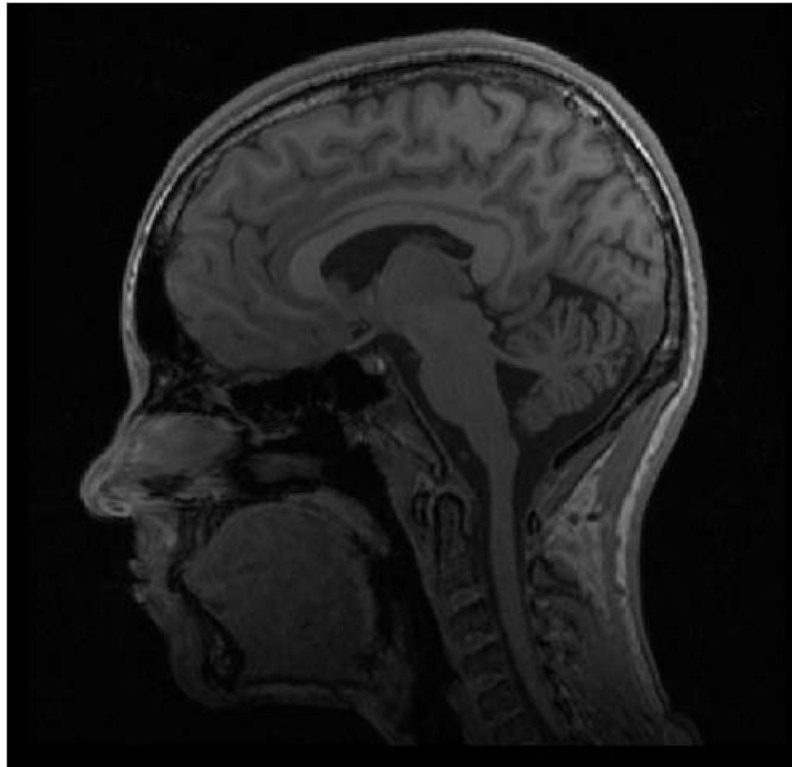


Credit: NASA, Jeff Hester, and Paul Scowen (Arizona State)
[More info here](#)

Applications of Image Processing



Medicine



Credit: Dr. Janet Lainhart, UofU Psychiatry

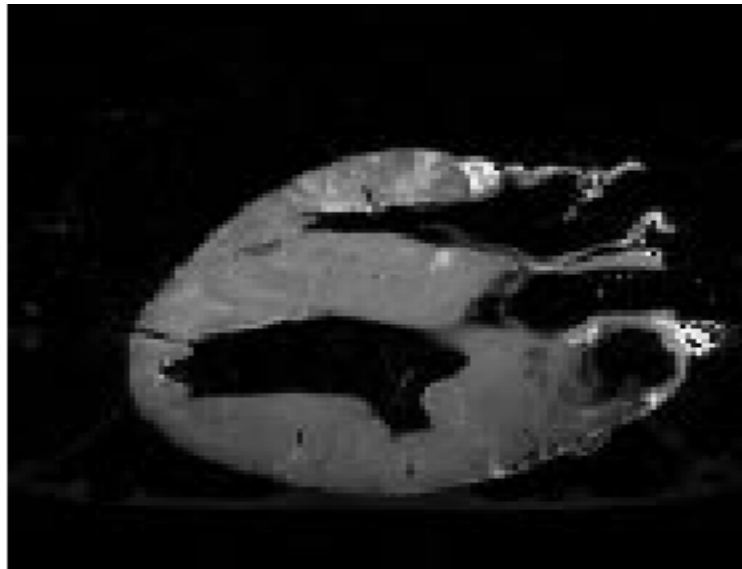
Security, Biometrics



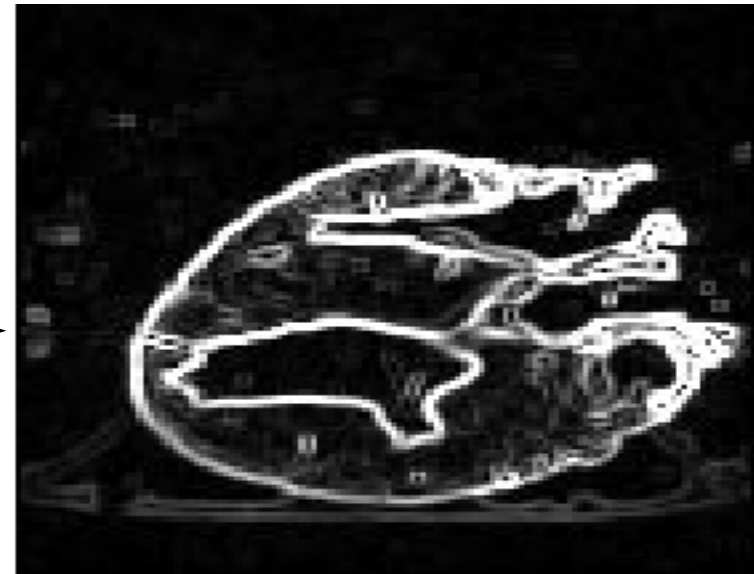


Applications of Image Processing: Medicine

Images taken from Gonzalez & Woods, Digital Image Processing (2002)



Original MRI Image of a Dog Heart



Edge Detection Image

Applications of Image Processing



Satellite Imagery



Credit: NASA

Personal Photos

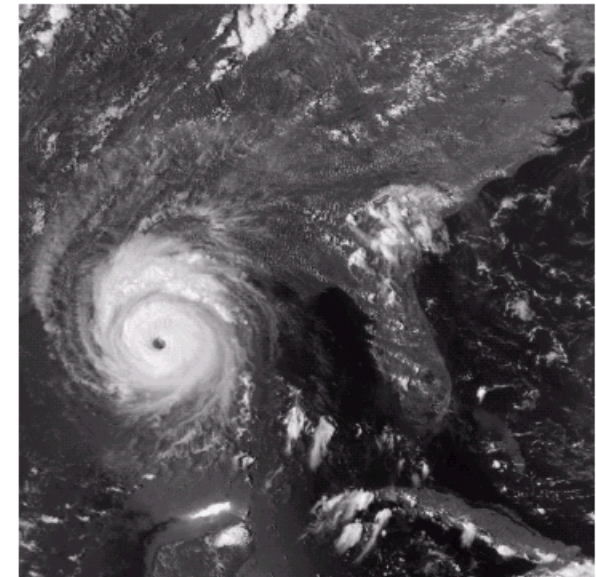
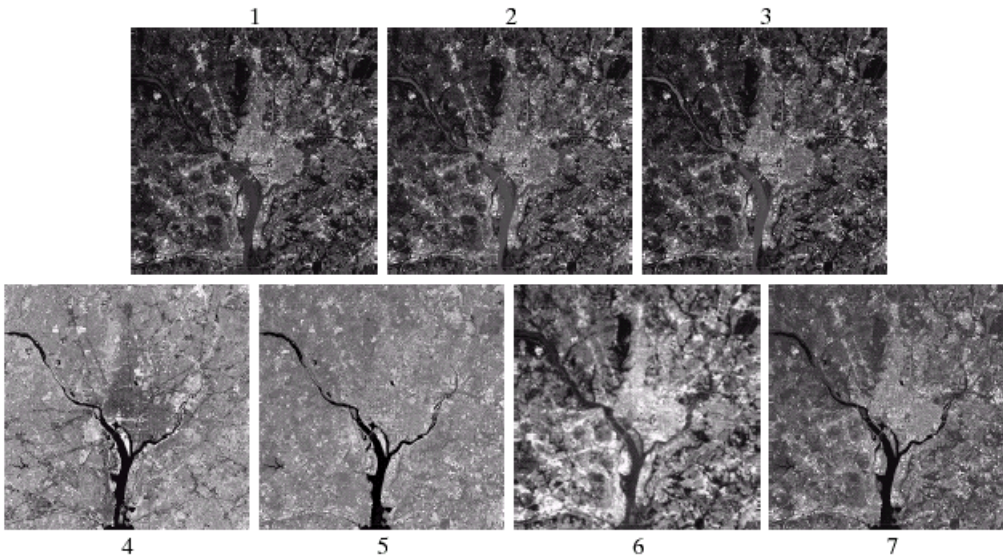


Credit: Tom Fletcher

Applications of Image Processing: Geographic Information Systems (GIS)



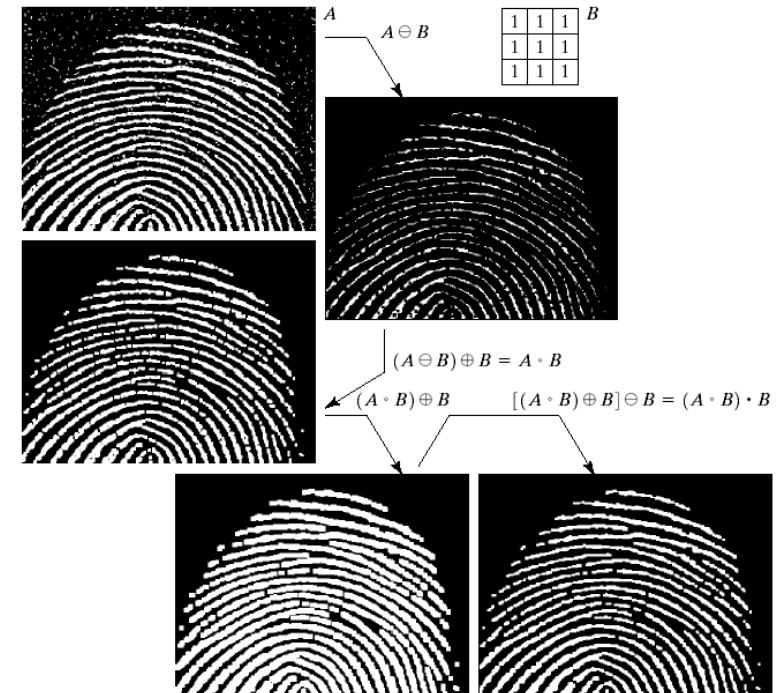
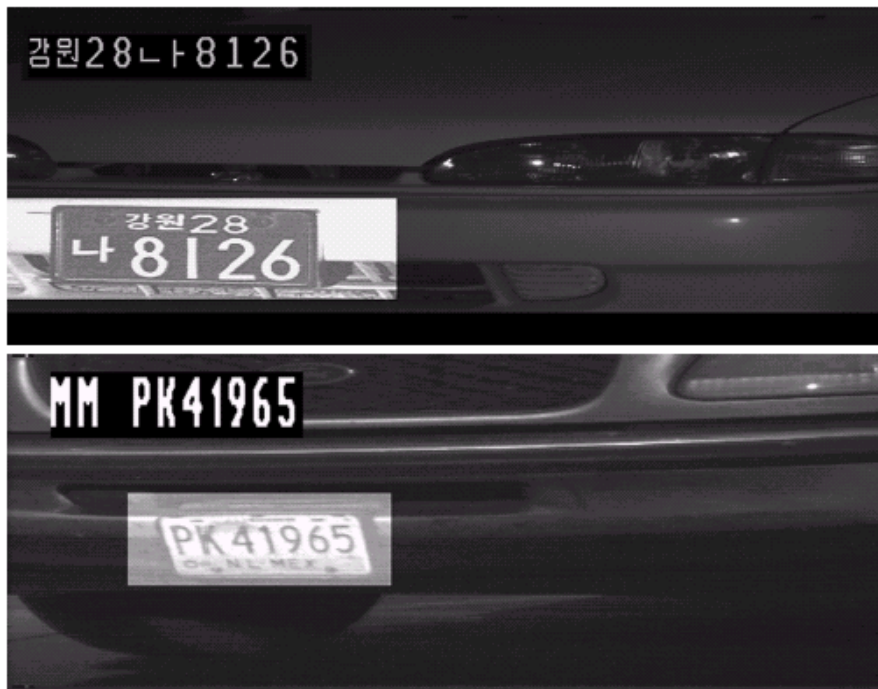
- Terrain classification
- Meteorology (weather)



Applications of Image Processing: Law Enforcement



- Number plate recognition for speed cameras or automated toll systems
- Fingerprint recognition



Applications of Image Processing: HCI



- Face recognition
- Gesture recognition





Relationship with other Fields

High-level

Computer Vision

Object detection, recognition, shape analysis, tracking
Use of Artificial Intelligence and Machine Learning

Image Analysis

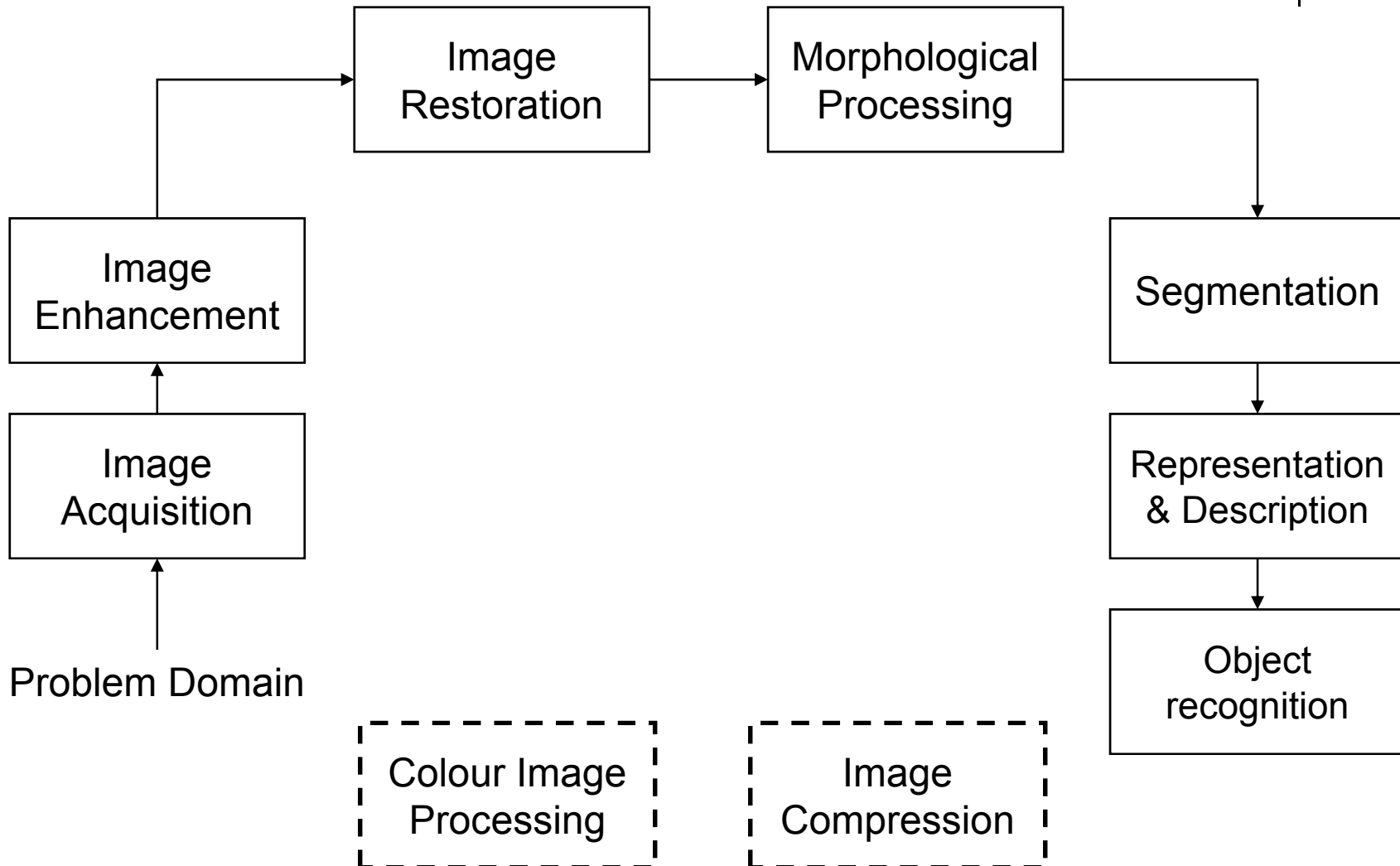
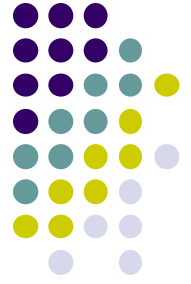
Segmentation, image registration, matching

Low-level

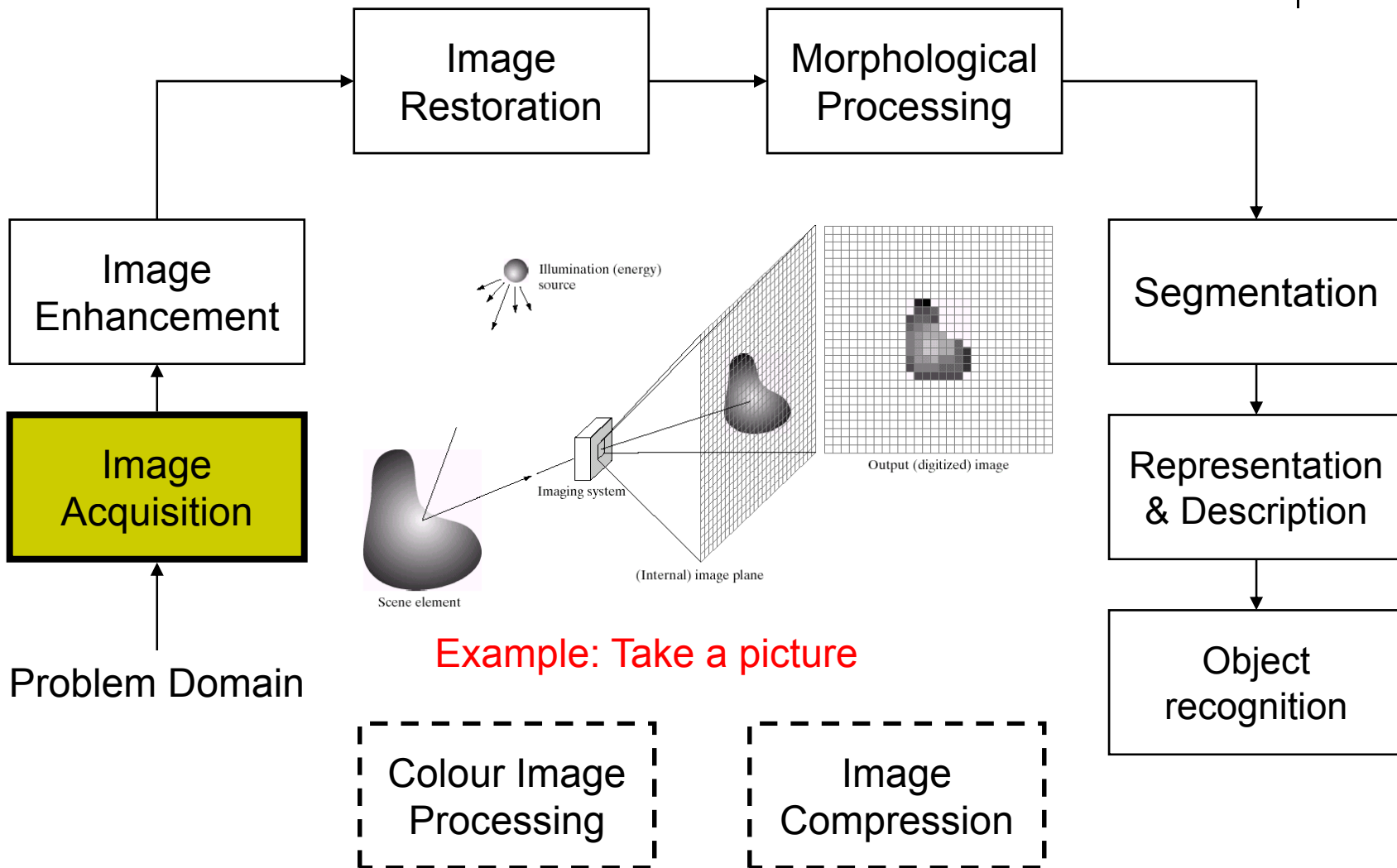
Image Processing

Image enhancement, noise removal, restoration,
feature detection, compression

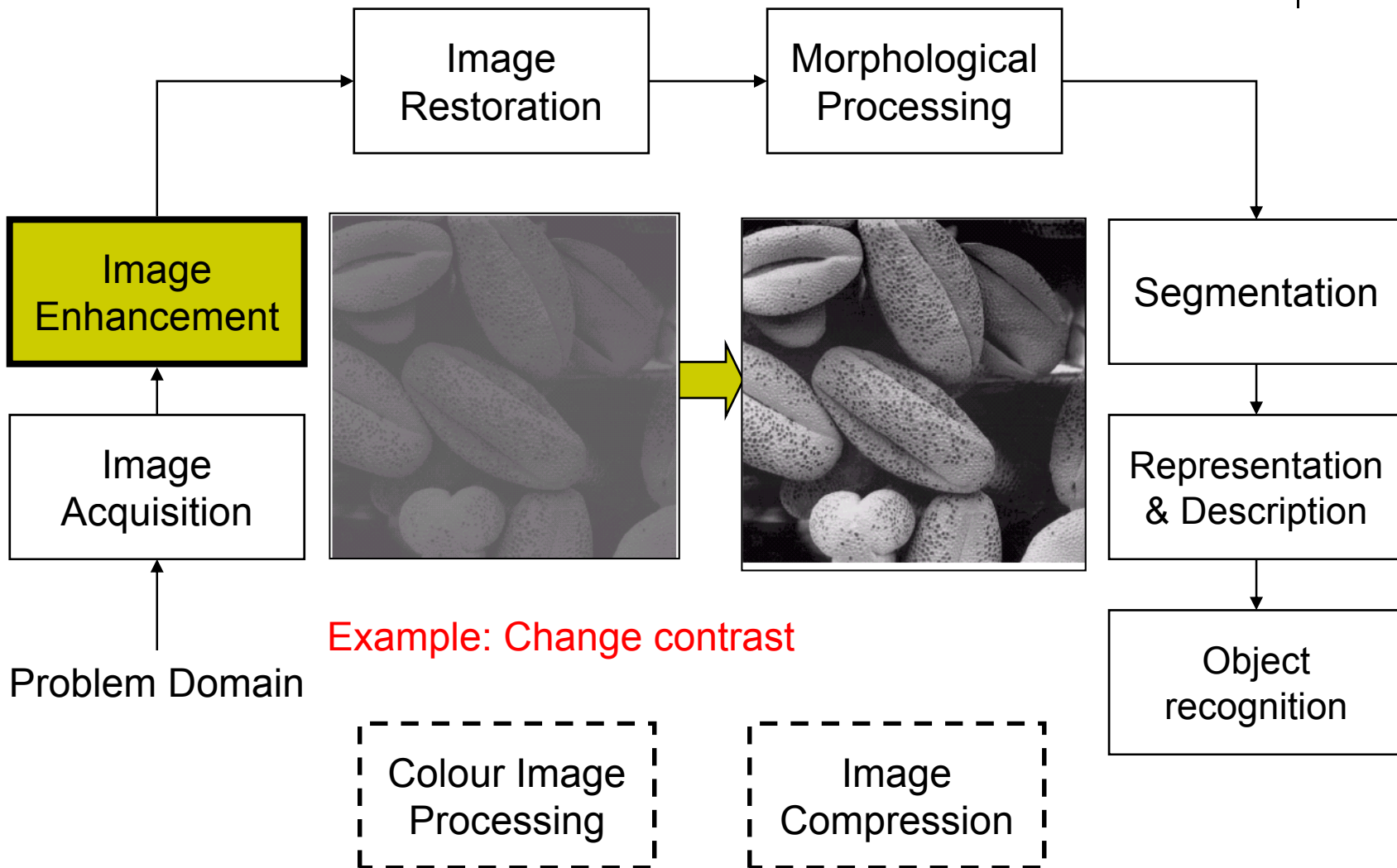
Key Stages in Digital Image Processing



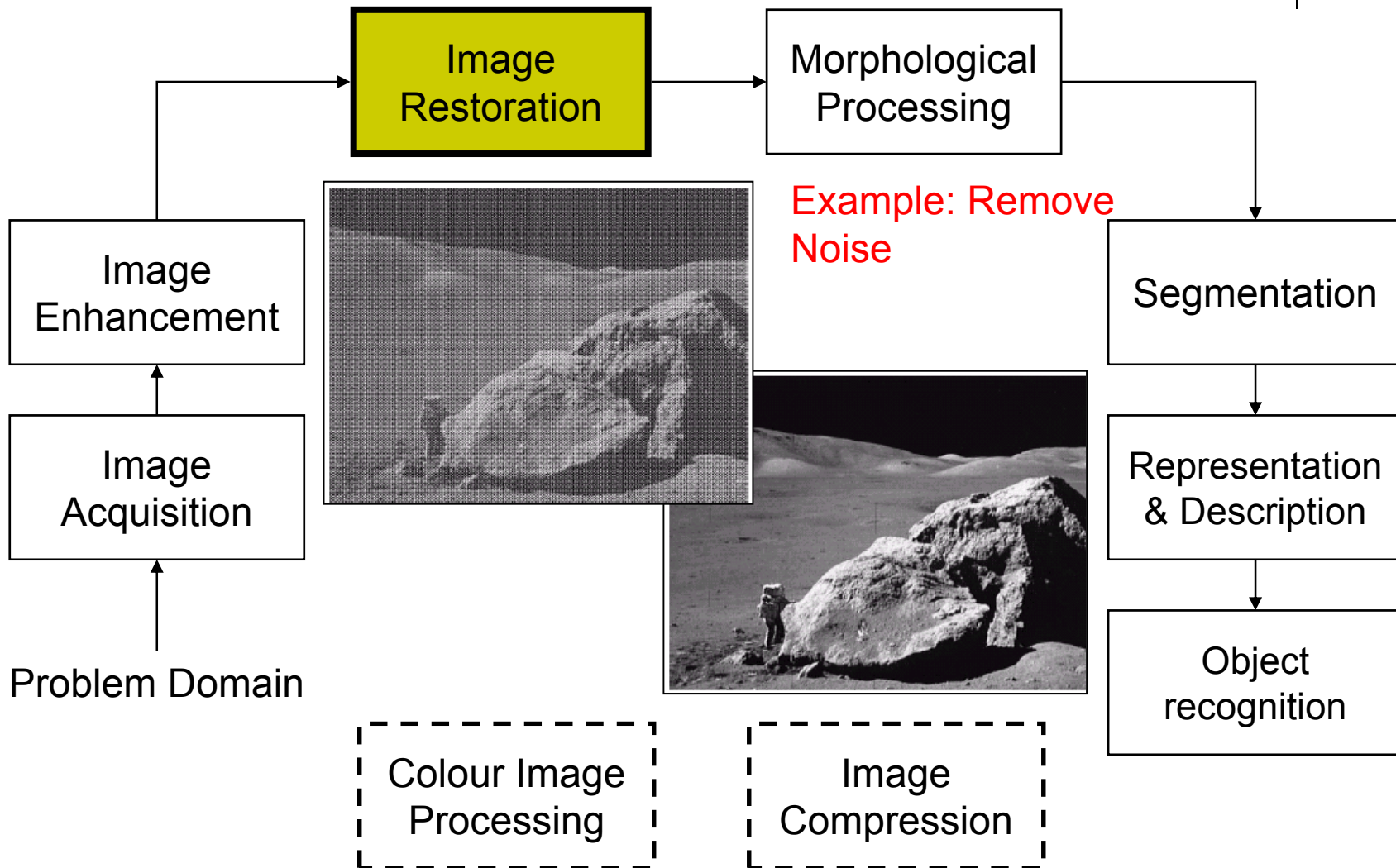
Key Stages in Digital Image Processing: Image Aquisition



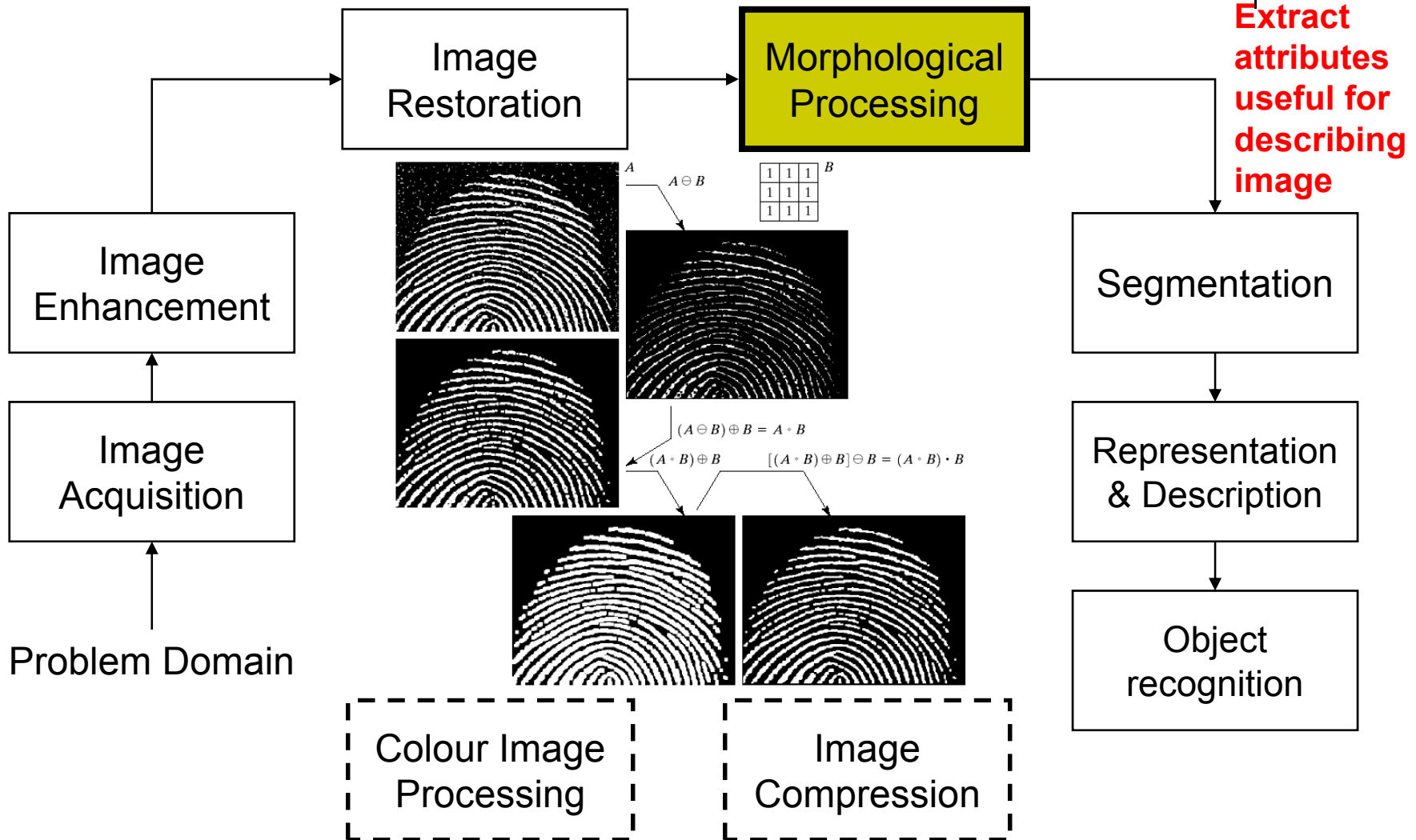
Key Stages in Digital Image Processing: Image Enhancement



Key Stages in Digital Image Processing: Image Restoration



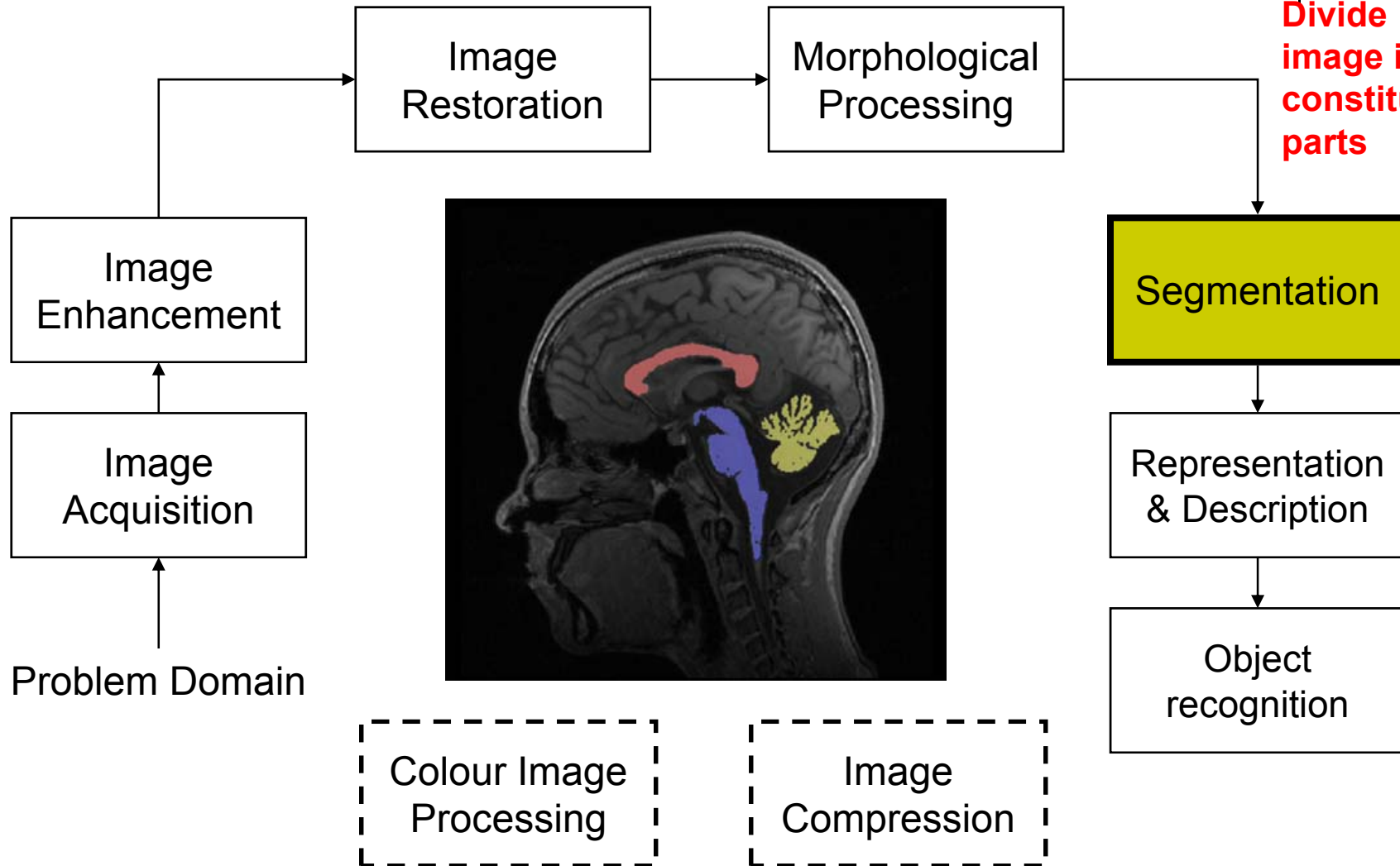
Key Stages in Digital Image Processing: Morphological Processing



Key Stages in Digital Image Processing: Segmentation



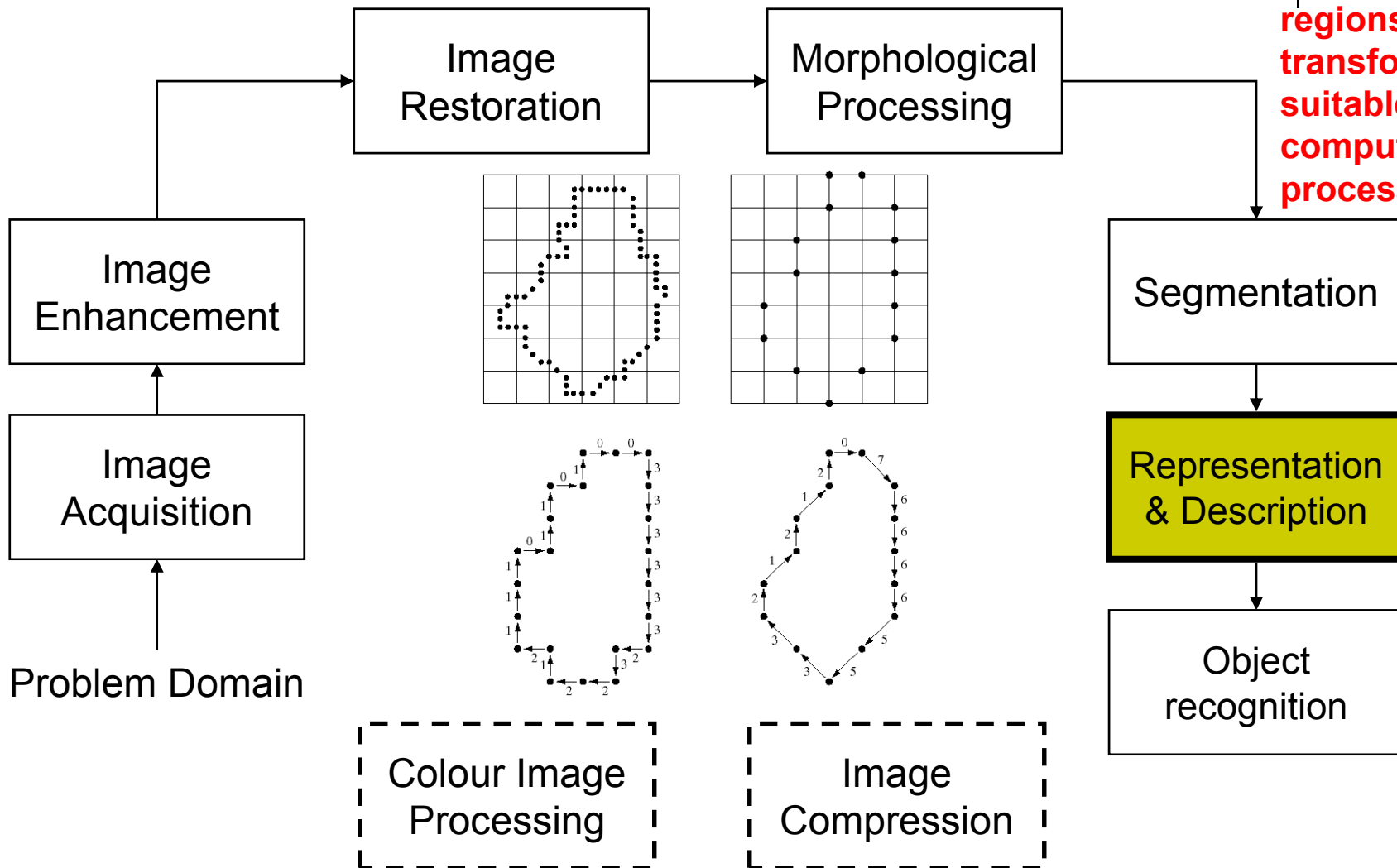
Divide
image into
constituent
parts



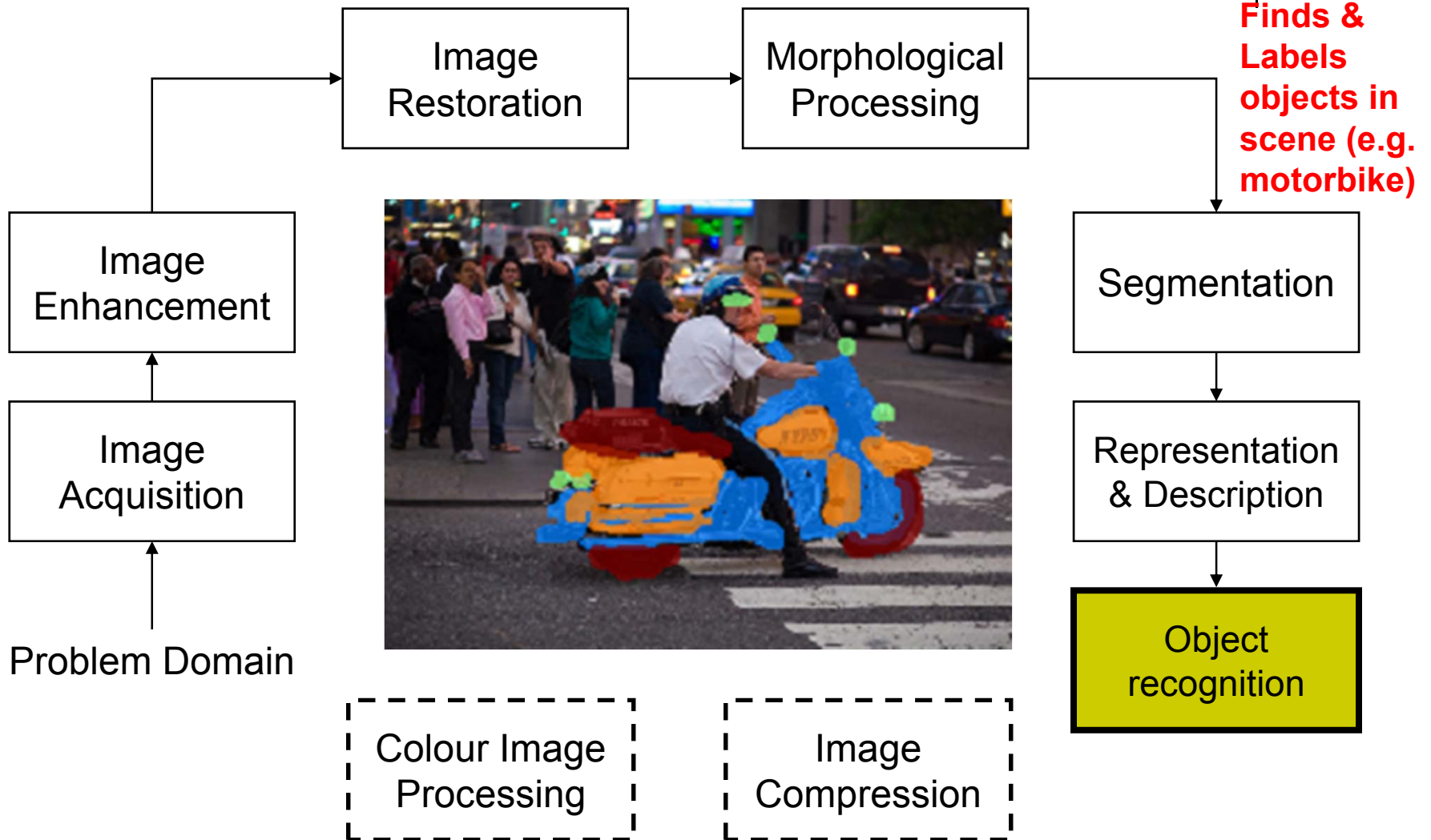
Key Stages in Digital Image Processing: Object Recognition

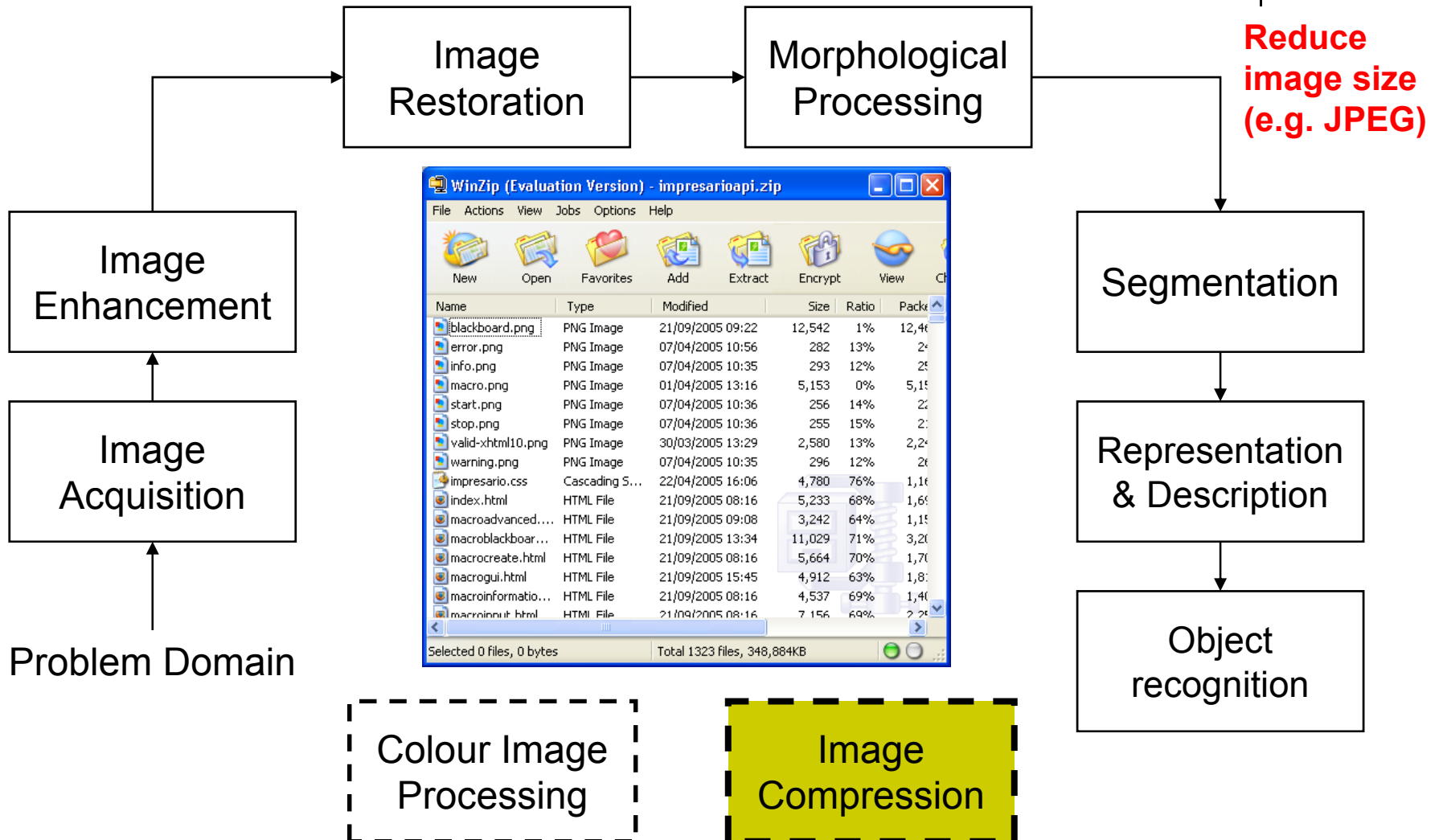


Image
regions
transformed
suitable for
computer
processing



Key Stages in Digital Image Processing: Representation & Description





Key Stages in Digital Image Processing: Colour Image Processing

